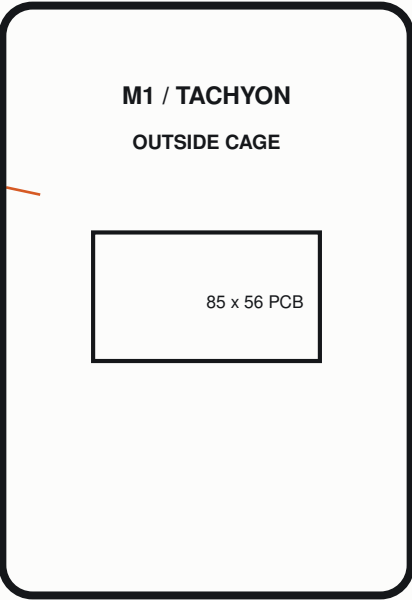
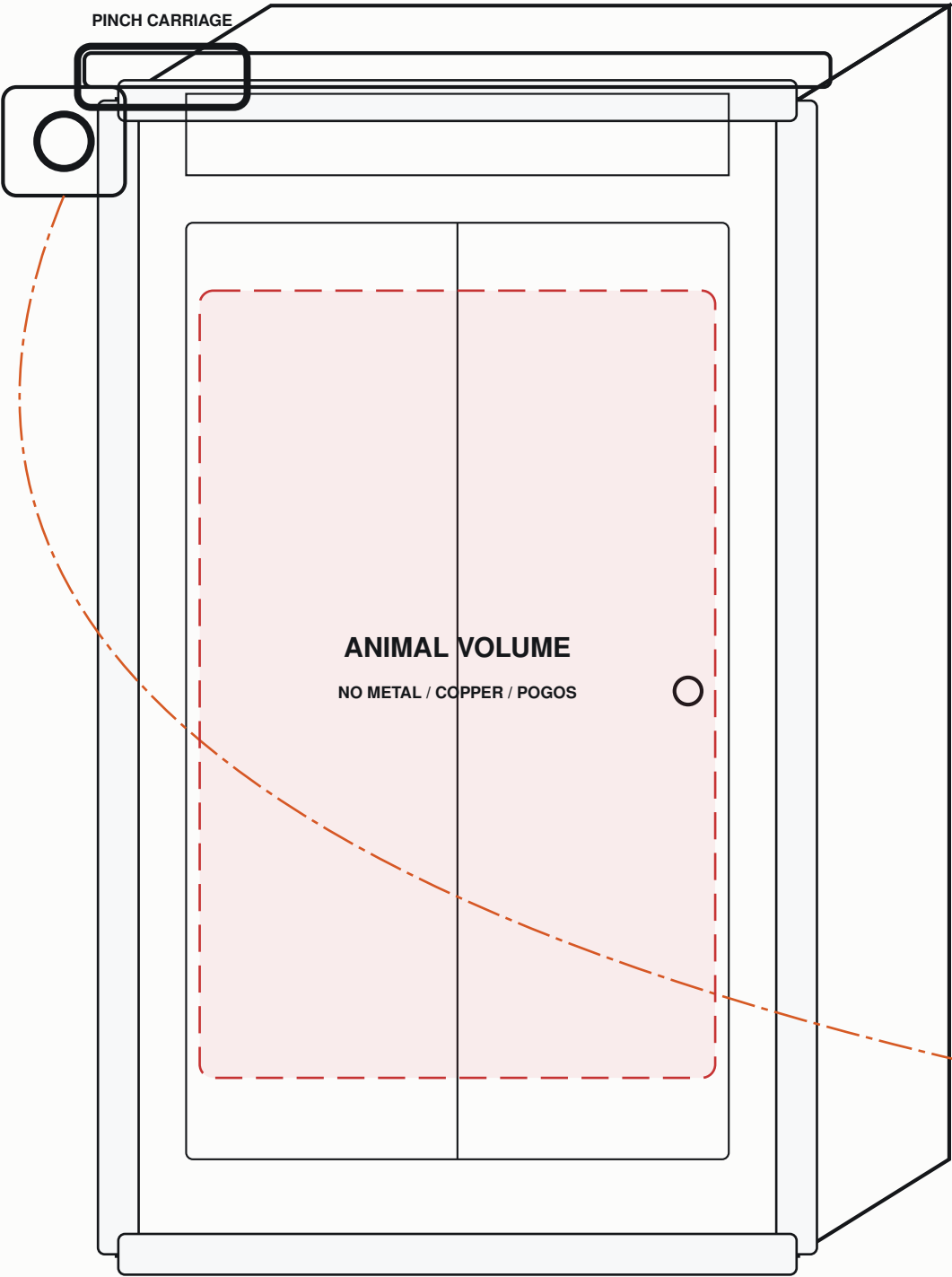


READABLE THEORY OF THE MACHINE

Exterior electrification; interior remains plastic, screened, and animal-safe.



SYSTEM THEORY

250 mm block perimeter; 500 mm first tower.
External rail carries fused VIN + I2C + ID.
Pinch lifts contacts before the carriage rolls.
One latch receives camera, sensor, LED, or mist binders.
Camera binder serializes CSI; GMSL2 video crosses locked pogo docks.
The M1 is a compute brick, never the animal cage.

LOCKED SAFETY BOUNDARY

Upper third remains free for hang-molt clearance.
Ceiling and vents use ≤ 0.8 mm non-metal screen.
B20 separates the guarded rail channel from the wet volume.
Door opens front/side; all gaps receive nymph check.
No taxonomy inferred: no live-mantis photo supplied.

DRAWING STATUS

Pitch / span / bus topology are source locks.
Named vendor capabilities are checked against primary docs.
Custom rail dimensions are REF/TARGET until tested.
Pogo series, strip stack and latch force UNVERIFIED.
Fabricate coupons before a live-animal first article.

NTS - DO NOT SCALE

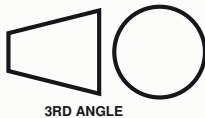
PARTICLE-BASE MANTIS TERRARIUM

COVER / THEORY

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



SCALE
NTS

DATE
2026-08-20

REV
B-DRAFT

SHEET
01 OF 12 / S00

STATUS
CONCEPT LOCK

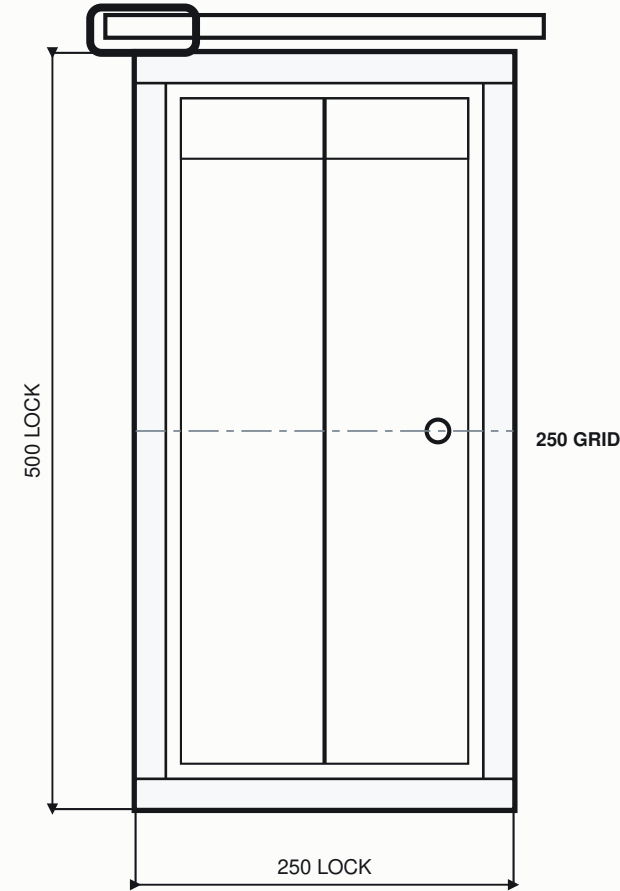
DRAWN
MANTIS LAB

DWG
MT-S00

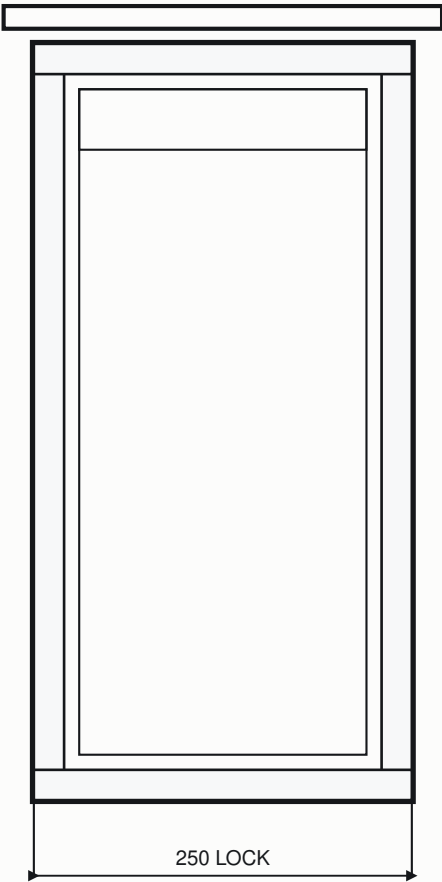
FORMAT
A3 SVG/PDF

TOL
AS NOTED

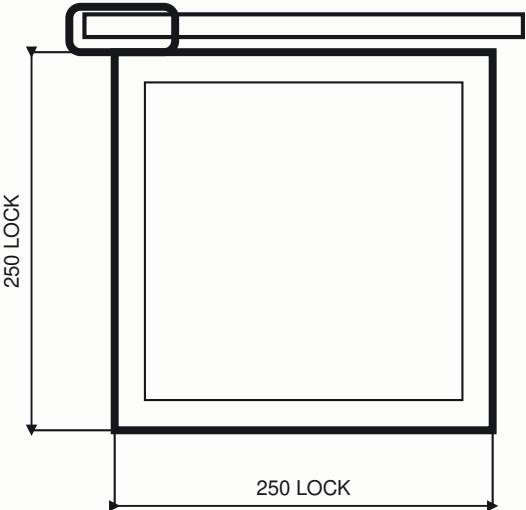
FRONT ELEVATION
SCALE NTS - DO NOT SCALE



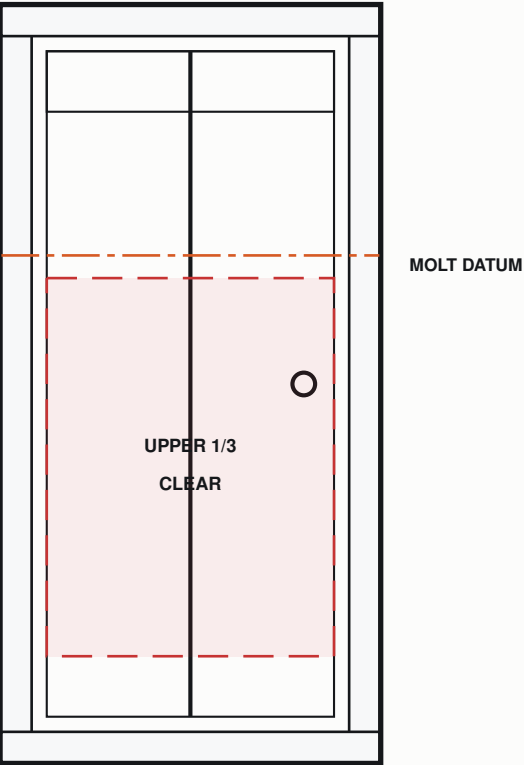
RIGHT ELEVATION
SCALE NTS - DO NOT SCALE



TOP PLAN
SCALE NTS - DO NOT SCALE



DATUM / KEEP-OUT
SCALE NTS - DO NOT SCALE



LOCKED ENVELOPE

Exterior: 250 W x 250 D x 500 H.
Module datum repeats at 250 mm.
Nominal clear: 202 x 202 x 427 REF.
Frame face dimensions REF pending first article.
Interior must still satisfy >=240 mm height.

PROJECTION

Third-angle arrangement shown.
Front is the door/view cassette face.
Top rail sits entirely outside enclosure.
Hidden wet-side barrier is continuous.

INTERFACE DATUMS

A: exterior block face.
B: cassette seat plane.
C: rail conductor centerline.
D: false-bottom tray surface.

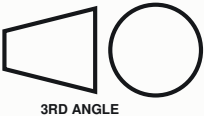
NTS - DO NOT SCALE

PARTICLE-BASE MANTIS TERRARIUM
ORTHOGRAPHIC ASSEMBLY

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

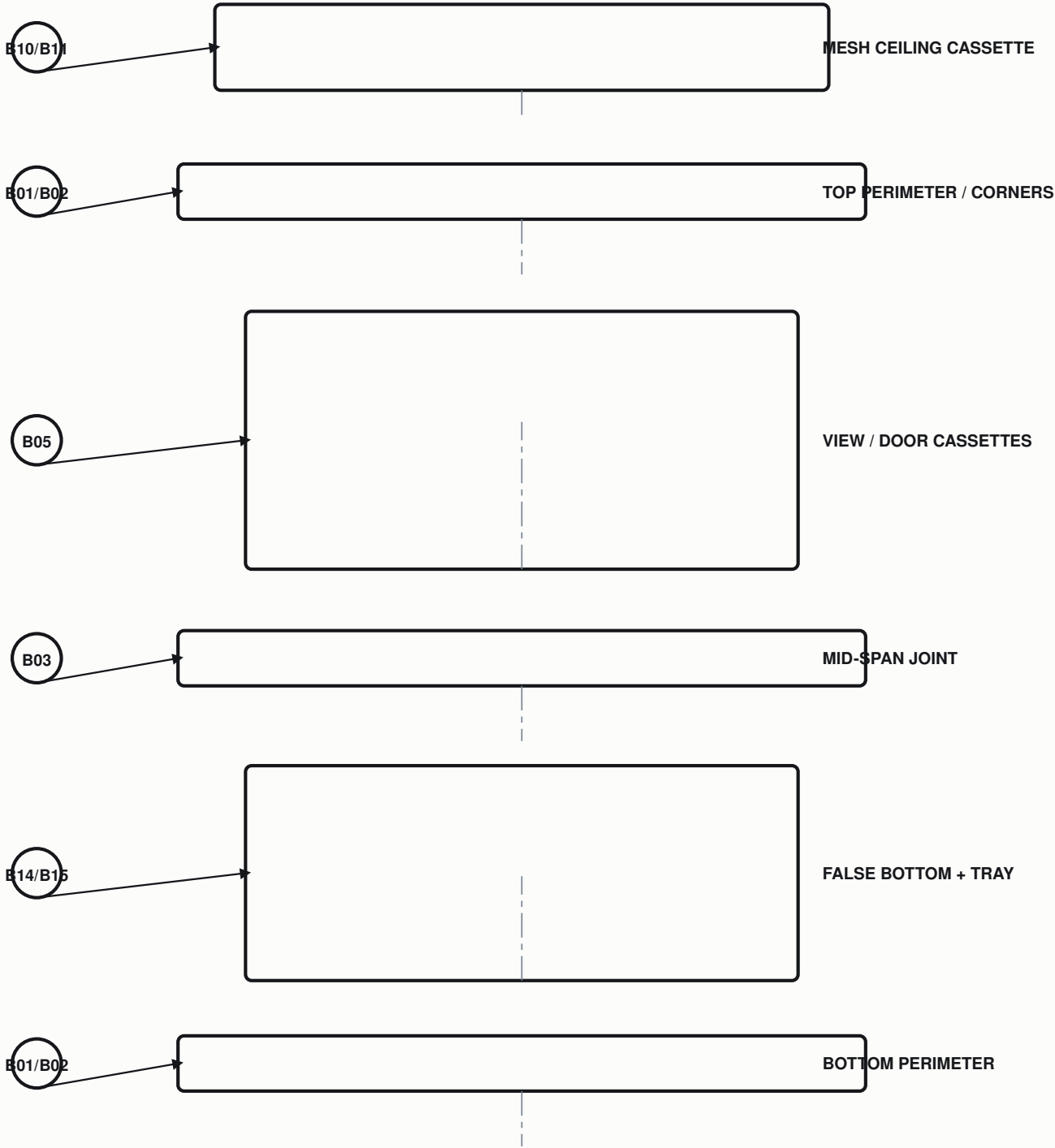
NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-DRAFT
SHEET 02 OF 12 / S01	STATUS CONCEPT LOCK	DRAWN MANTIS LAB
DWG MT-S01	FORMAT A3 SVG/PDF	TOL AS NOTED

FUNCTIONAL-AXIS EXPLOSION

SCALE NTS - DO NOT SCALE

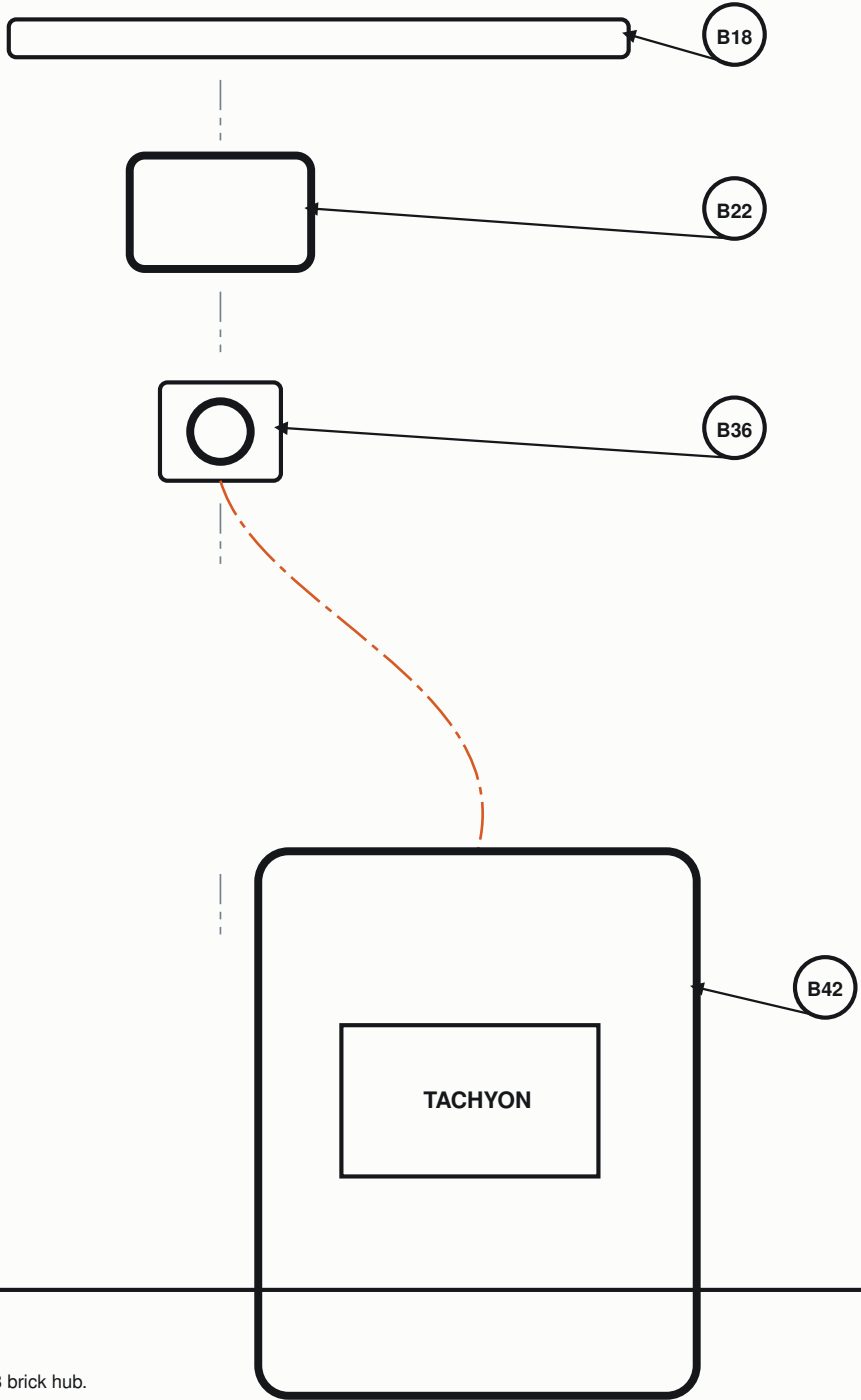


EXPLOSION RULE

Subassemblies remain grouped; trail lines follow service axes.
Rail, carriage, binder and compute brick are removable externally.
No service operation opens a path from rail metal into wet volume.

INSTALL ORDER

1 perimeter -> 2 faces -> 3 ceiling -> 4 false bottom.
5 animal-isolated rail -> 6 carriage -> 7 SerDes binder -> 8 brick hub.
First article requires leak, pinch, pull-off and nymph-gap checks.

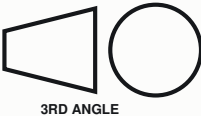


PARTICLE-BASE MANTIS TERRARIUM
EXPLODED ASSEMBLY

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

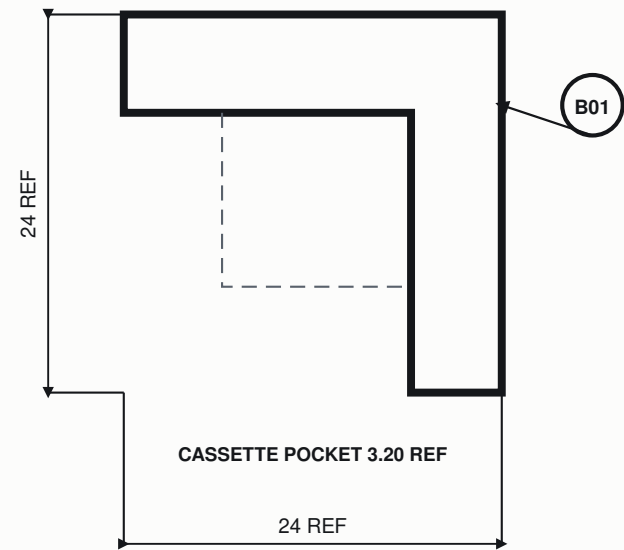
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE

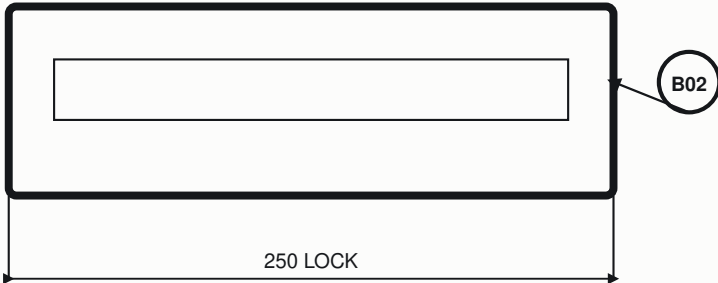


SCALE NTS	DATE 2026-08-20	REV B-DRAFT
SHEET 03 OF 12 / S02	STATUS CONCEPT LOCK	DRAWN MANTIS LAB
DWG MT-S02	FORMAT A3 SVG/PDF	TOL AS NOTED

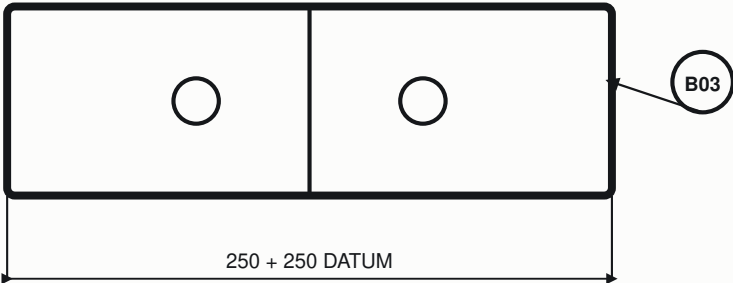
CORNER BLOCK
SCALE NTS - DO NOT SCALE



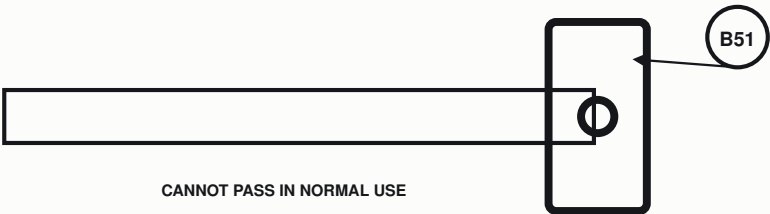
EDGE BLOCK
SCALE NTS - DO NOT SCALE



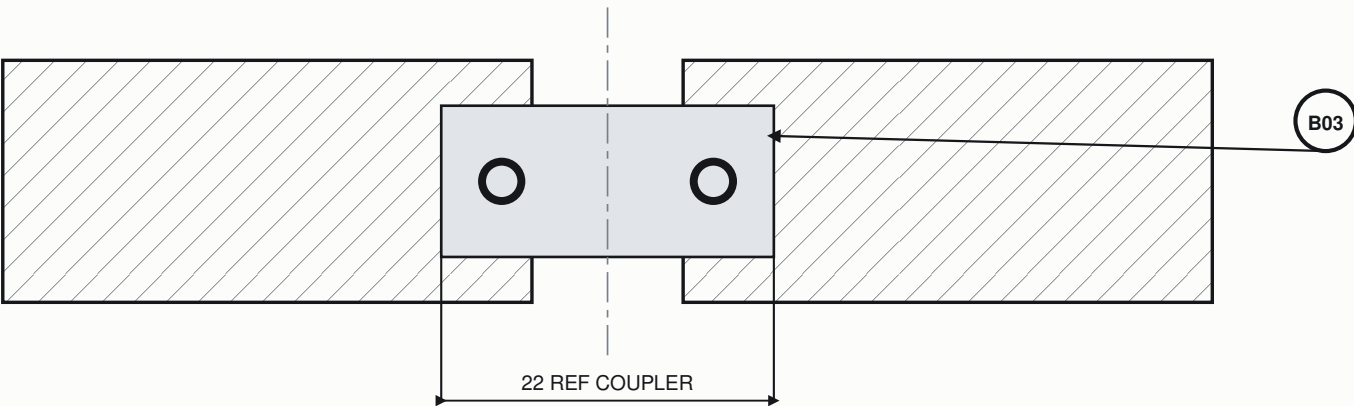
MID-SPAN BLOCK
SCALE NTS - DO NOT SCALE



ROUTE END STOP / SERVICE REMOVAL
SCALE NTS - DO NOT SCALE

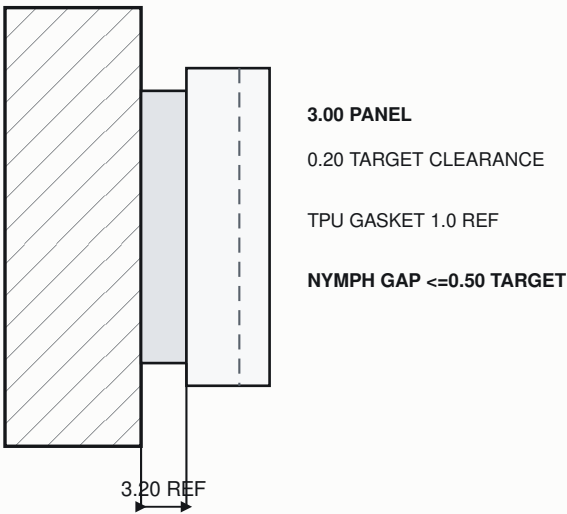


DETAIL A - 250 -> 500 SPAN JOINT
SCALE NTS - DO NOT SCALE



M3 HEAT-SET + CROSS DOWEL; SHEAR CARRIED BY TONGUE

DETAIL B - CASSETTE SEAT
SCALE NTS - DO NOT SCALE



MECHANICAL STATUS

Block cross-sections are REF: no commercial-frame photo supplied.
Coupon the cassette pocket at 3.10 / 3.20 / 3.30 before release.

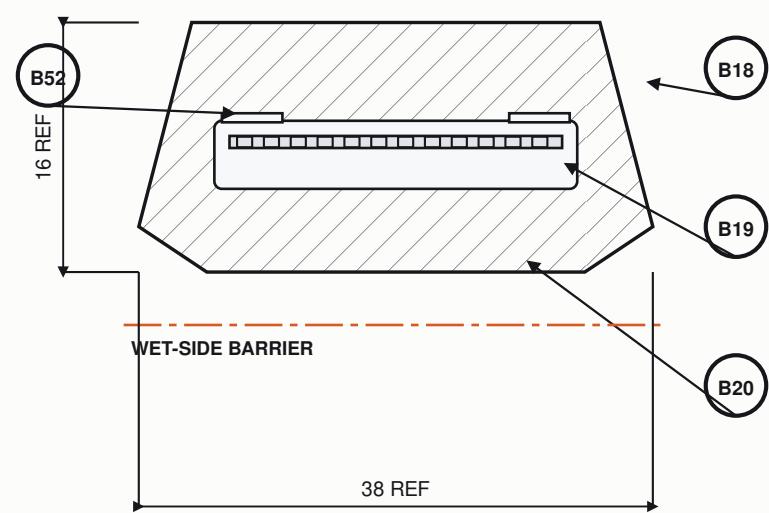
LOAD PATH

Panel pressure -> gasket -> pocket wall -> perimeter block.
500 span bending crosses the internal coupler, not the face cassette.
V1 rail corners are electrical junctions; carriage does not roll around 90 deg.
B51 positively retains each route; removal needs deliberate M3 service.

SCALE NTS	DATE 2026-08-20	REV B-DRAFT
SHEET 04 OF 12 / S03	STATUS CONCEPT LOCK	DRAWN MANTIS LAB
DWG MT-S03	FORMAT A3 SVG/PDF	TOL AS NOTED

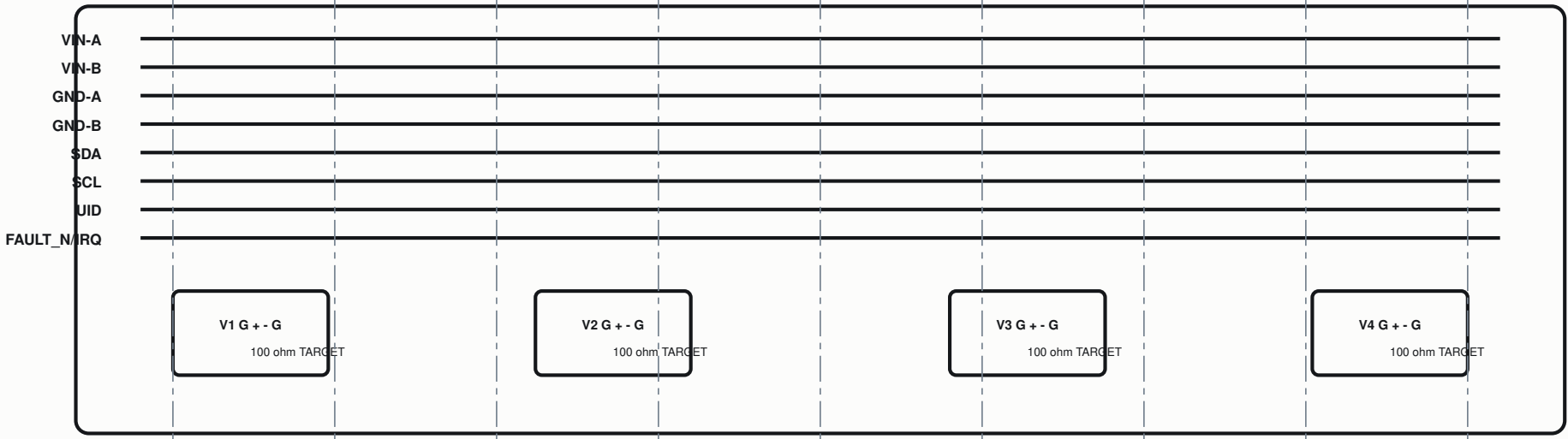
RAIL CHANNEL SECTION C-C

SCALE NTS - DO NOT SCALE



LONGITUDINAL CONDUCTOR RUN

SCALE NTS - DO NOT SCALE

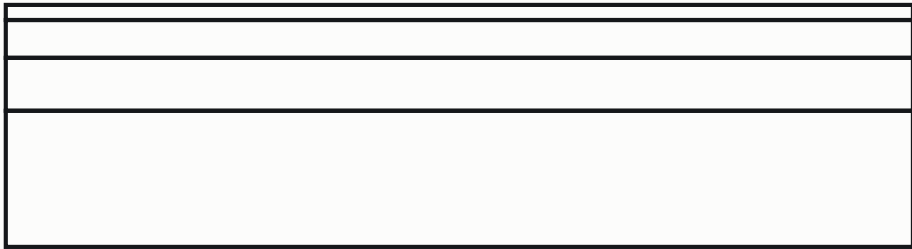


25 mm MECH DETENTS; VIDEO ONLY AT V1-V4 POINT-TO-POINT DOCKS

DETAIL D - STRIP STACK-UP

SCALE NTS - DO NOT SCALE

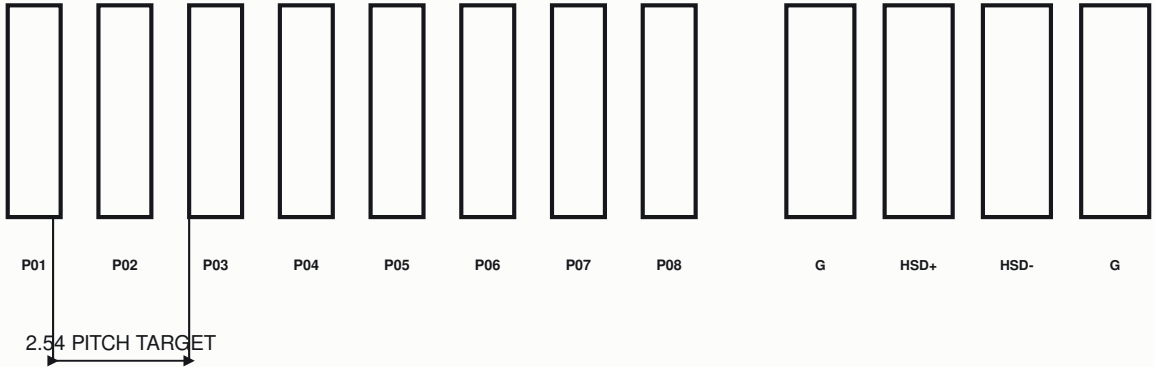
EXPOSED ENIG LAND IN CAPTIVE EXTERNAL SLOT - NO FILM OVER CONTACT



ALL THICKNESSES UNVERIFIED UNTIL STRIP VENDOR SELECTED

DETAIL E - CONTACT LANDS

SCALE NTS - DO NOT SCALE



P01-P08 CONTINUOUS; P09-P12 GUARDED GMSL2 DOCK CELL

POWER / SIGNAL

VIN: 12 V nominal TARGET from separate DC/DC; fuse F1 = 2 A TARGET.
P01-P08: power/control. P09-P12: 100 ohm TARGET GMSL2 cell.
Video contact exists only at V1-V4; no signal is carried while rolling.
B20 is the continuous wet-side barrier; contact lands stay external.

FAILURE CONTROL

S1 pre-travel opens the local carriage Q1 branch before pogo lift.
Bent HSD pogo -> link loss; no fallback to raw MIPI or continuous strip.
Each V-dock routes point-to-point to a separate deserializer input.
B52 guard/wiper shields the access slot; it is not a hermetic cover.

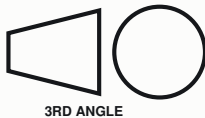
PARTICLE-BASE MANTIS TERRARIUM

RAIL + STRIP

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



3RD ANGLE

SCALE
NTS

SHEET
05 OF 12 / S04

DWG
MT-S04

DATE
2026-08-20

STATUS
CONCEPT LOCK

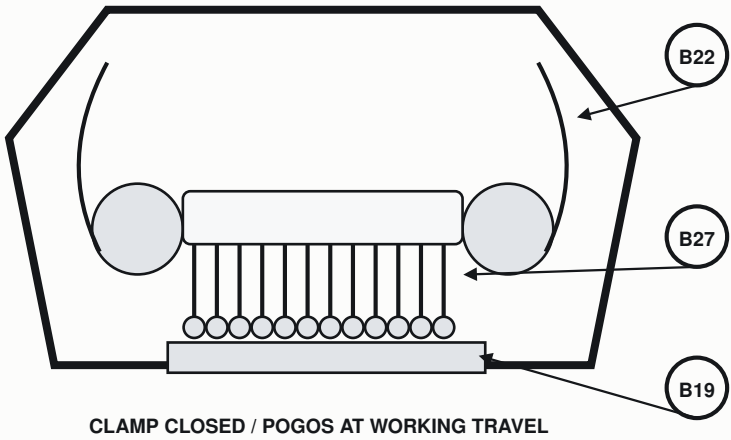
FORMAT
A3 SVG/PDF

REV
B-DRAFT

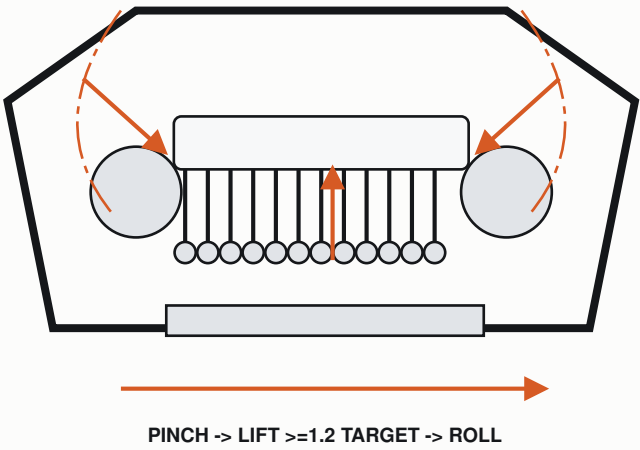
DRAWN
MANTIS LAB

TOL
AS NOTED

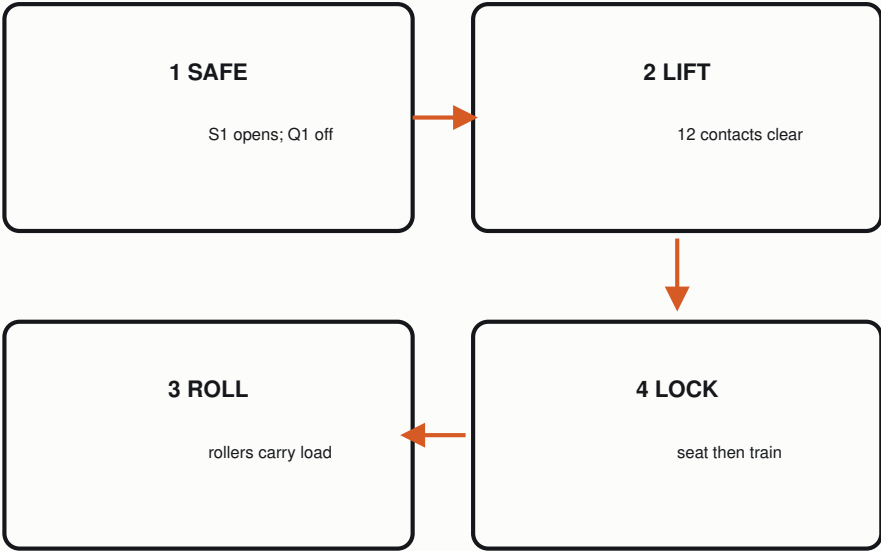
LOCKED STATE
SCALE NTS - DO NOT SCALE



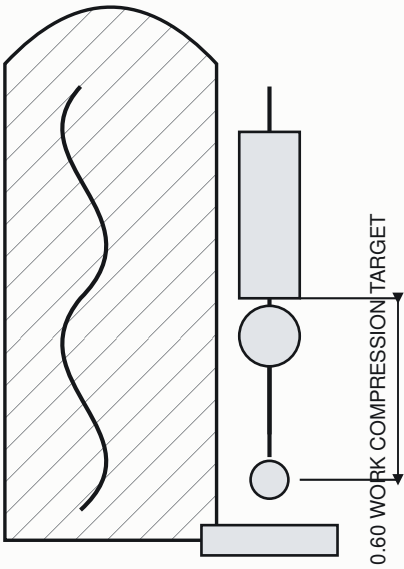
PINCHED / TRAVEL STATE
SCALE NTS - DO NOT SCALE



KINEMATIC SEQUENCE
SCALE NTS - DO NOT SCALE



DETAIL F - PINCH / SPRING / POGO ARRAY
SCALE NTS - DO NOT SCALE



SPRING: SELECT AFTER FORCE TEST

PINCH FORCE 15-25 N TARGET

12-CONTACT ARRAY; HSD CELL SI UNVERIFIED

ROLLER AXLE: M3 REF

DOVETAIL CLEARANCE 0.25/side TARGET

MECHANISM NOTE - PINCH-LIFT / ROLL

States: locked / pinched. Housing and load move; rail is grounded.
Local S1 opens Q1 and isolates signals before cam lifts contacts.
Cam lift must exceed pogo overtravel before rollers translate.
Bent HSD pogo: bounded training times out; local Q1 latches off.
Release seats clamp and binder, then a bounded link-training window.
Chosen over screws for one-hand relocation without opening cage.

ACCEPTANCE TARGETS

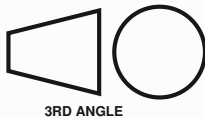
Clamp slip: <0.5 mm under 0.5 N-m binder moment.
Release cycles: 10,000 TARGET; contact resistance vendor-dependent.
B51 must retain a loaded carriage under the defined handling/drop test.
No conductive part may breach the wet-side barrier.

PARTICLE-BASE MANTIS TERRARIUM
CARRIAGE MECHANISM

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

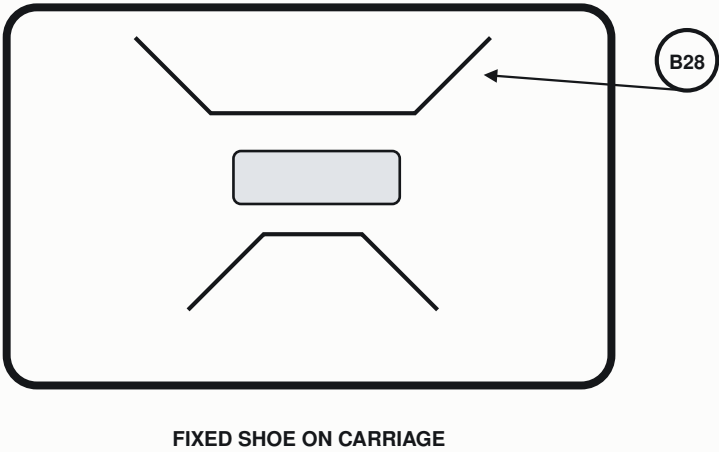
NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-DRAFT
SHEET 06 OF 12 / S05	STATUS CONCEPT LOCK	DRAWN MANTIS LAB
DWG MT-S05	FORMAT A3 SVG/PDF	TOL AS NOTED

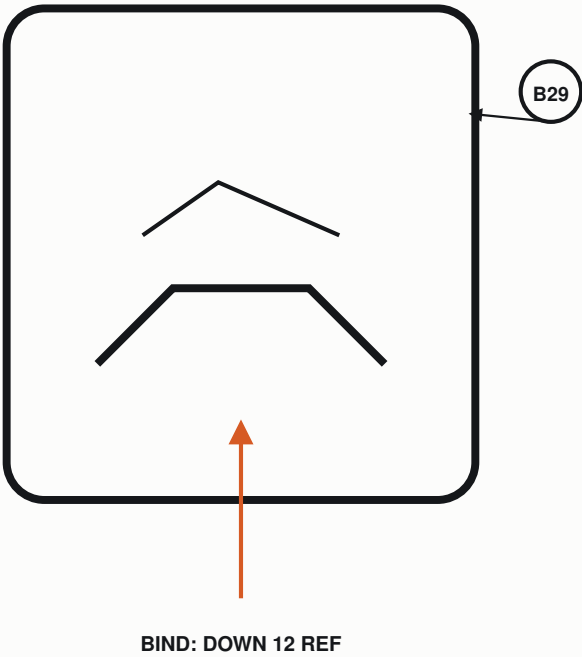
UNIVERSAL SHOE - FREE

SCALE NTS - DO NOT SCALE



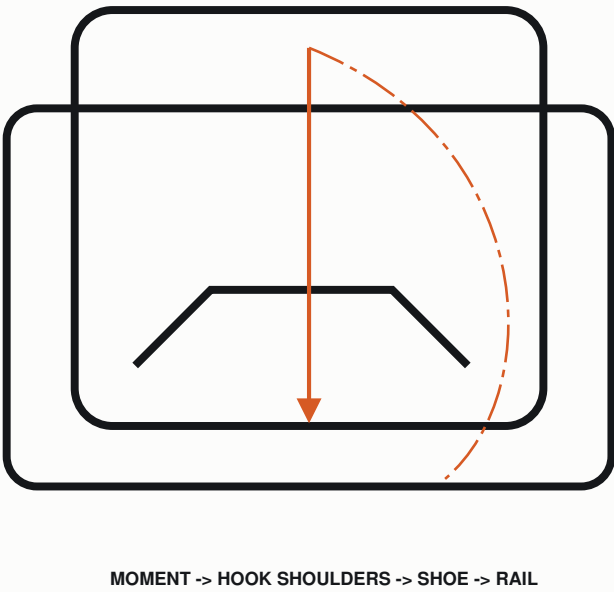
BINDER - MATING

SCALE NTS - DO NOT SCALE



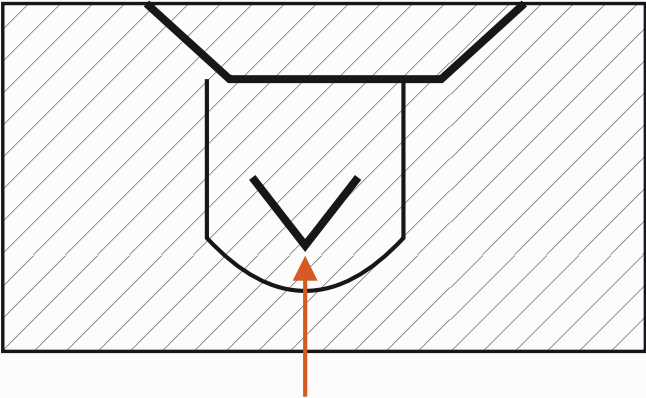
MATED / LOAD PATH

SCALE NTS - DO NOT SCALE



DETAIL G - CLICK LATCH SECTION

SCALE NTS - DO NOT SCALE



SPRING TONGUE DEFLECTION 1.2 TARGET

PULL-OFF ≥ 40 N TARGET

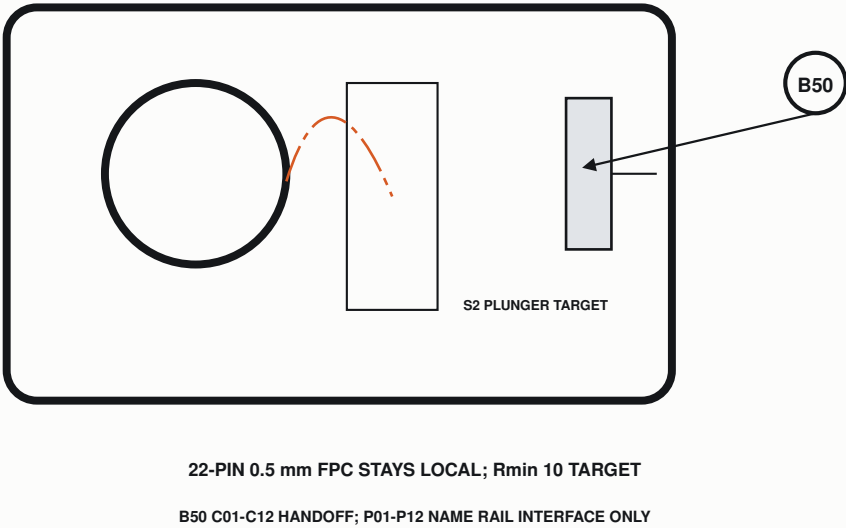
SIDE PLAY ≤ 0.20 TARGET

MATERIAL: PETG/ASA BODY + POM WEAR KEY REF

FAILURE: CRACK / PARTIAL ENGAGEMENT -> RETENTION WITNESS MARK

DETAIL H - BINDER-LOCAL FPC + B50 FACE

SCALE NTS - DO NOT SCALE



MECHANISM NOTE - BIND-LATCH

States: pinch-safe / free / mated. Binder moves; shoe stays on carriage.

Tongue snaps behind shoulder; S2 confirms full electrical seating.

Release is mechanically blocked unless carriage is PINCH-SAFE.

Chosen over a screw for tool-free swap without disturbing rail position.

FAILURE / CLEARANCE

Partial click: S2 stays open and local Q1 must remain off.

B50 series, mate order, SI launch and hot-unplug timing are UNVERIFIED.

Internal camera-to-serializer FPC never carries binder moment.

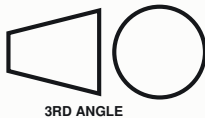
PARTICLE-BASE MANTIS TERRARIUM

UNIVERSAL LATCH + BINDER

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

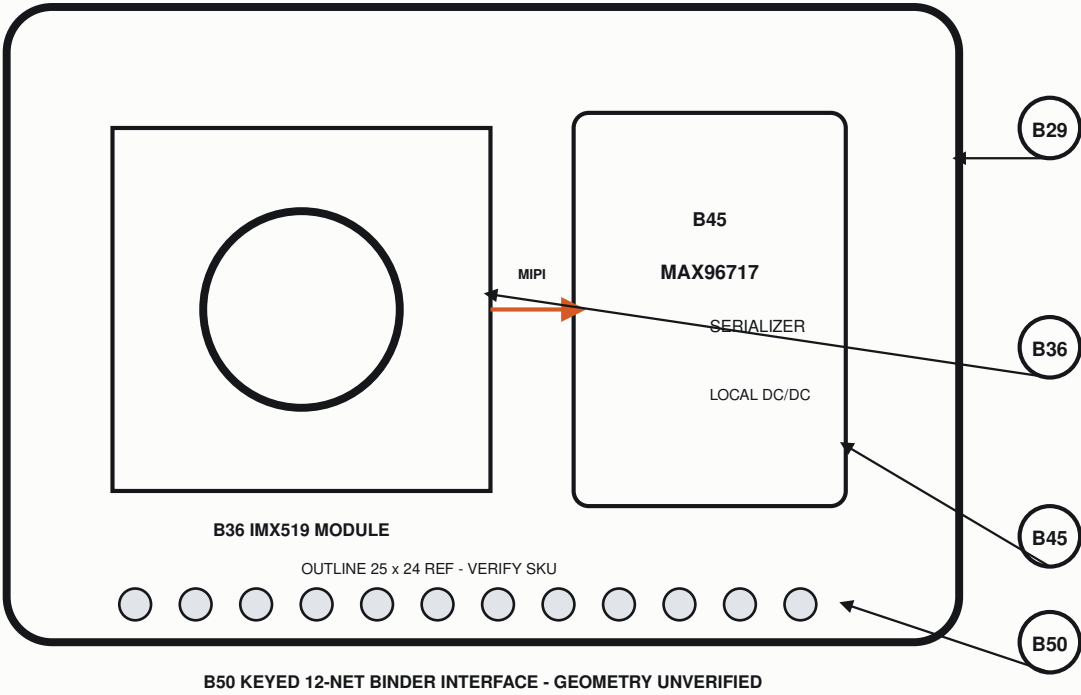
NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-DRAFT
SHEET 07 OF 12 / S06	STATUS CONCEPT LOCK	DRAWN MANTIS LAB
DWG MT-S06	FORMAT A3 SVG/PDF	TOL AS NOTED

CAMERA BINDER ASSEMBLY

SCALE NTS - DO NOT SCALE



RAIL PXX / BINDER CXX MIRRORED NETS

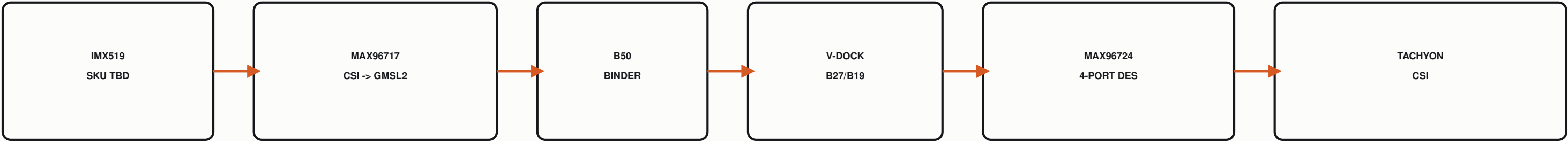
SCALE NTS - DO NOT SCALE



B27 + B50 FULL CHANNEL: 100 ohm TARGET / TO BE CHARACTERIZED

VIDEO PATH - NO TETHER TO MOVING CARRIAGE

SCALE NTS - DO NOT SCALE



RAW MIPI STAYS INSIDE BINDER; 3/6 Gbps GMSL2 CROSSES LOCKED POGO DOCK

POWER + VIDEO + REVERSE CONTROL CROSS THE CARRIAGE; NOTHING TETHERS IT

MECHANISM NOTE - CAMERA VIDEO DOCK

States: free / S1+S2 power-mated / training / link-trained / pinch-safe.
Binder moves; V-dock and deserializer are fixed to the rail/brick.
S1 or S2 opening mutes link and turns local Q1 off before contact motion.

FAILURE / WHY

Partial dock or bent HSD pin -> bounded training timeout; Q1 latches off.
Chosen over a CSI tether so the carriage remains genuinely repositionable.
Coupon and eye/BER tests must include both B27 and B50 interfaces.

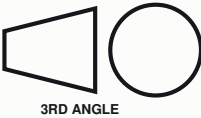
PARTICLE-BASE MANTIS TERRARIUM

CAMERA LOAD / SERDES BINDER

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

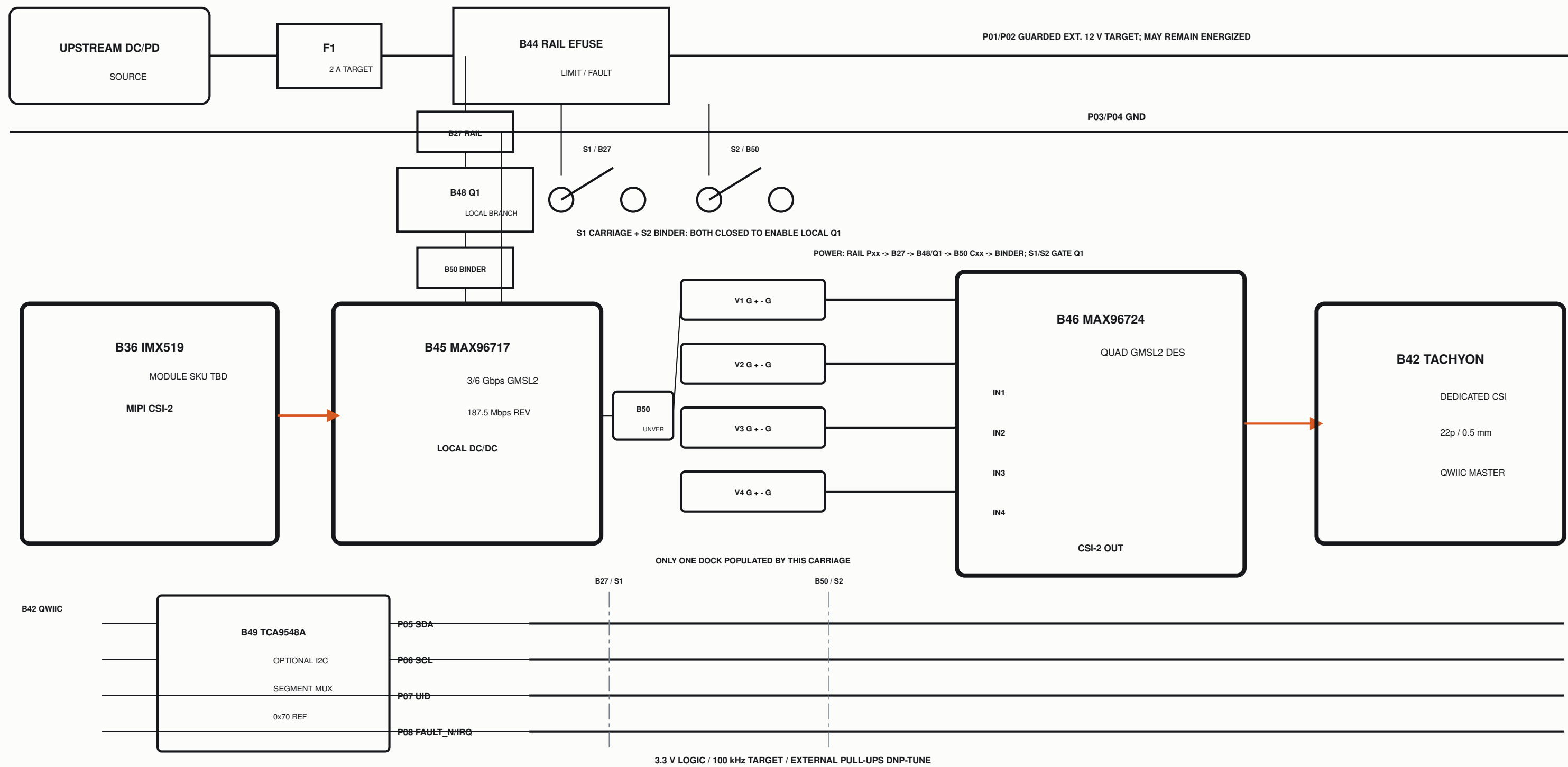
Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-DRAFT
SHEET 08 OF 12 / S07	STATUS CONCEPT LOCK	DRAWN MANTIS LAB
DWG MT-S07	FORMAT A3 SVG/PDF	TOL AS NOTED

RAIL POWER / CONTROL / GMSL2 - FUNCTIONAL DIAGRAM, NOT RELEASE SCHEMATIC



MECHANISM NOTE - PROTECTED VIDEO MATE

States: absent / sealed / training / trained / fault / pinch-safe / lifted.
S1 or S2 opens first; local Q1 turns off and discharges before motion.
Release seats B27+B50 before a bounded, current-limited training window.

FAULT POLICY

Bent HSD pin or station skew -> no lock; training times out and Q1 latches off.
 UID identifies load, while V1-V4 deserializer port identifies position.
 P08 is diagnostic only. B27+B50 eye/BER, ESD and timing remain gates.

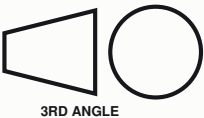
PARTICLE-BASE MANTIS TERRARIUM

ELECTRICAL + VIDEO FUNCTIONAL DIAGRAM

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

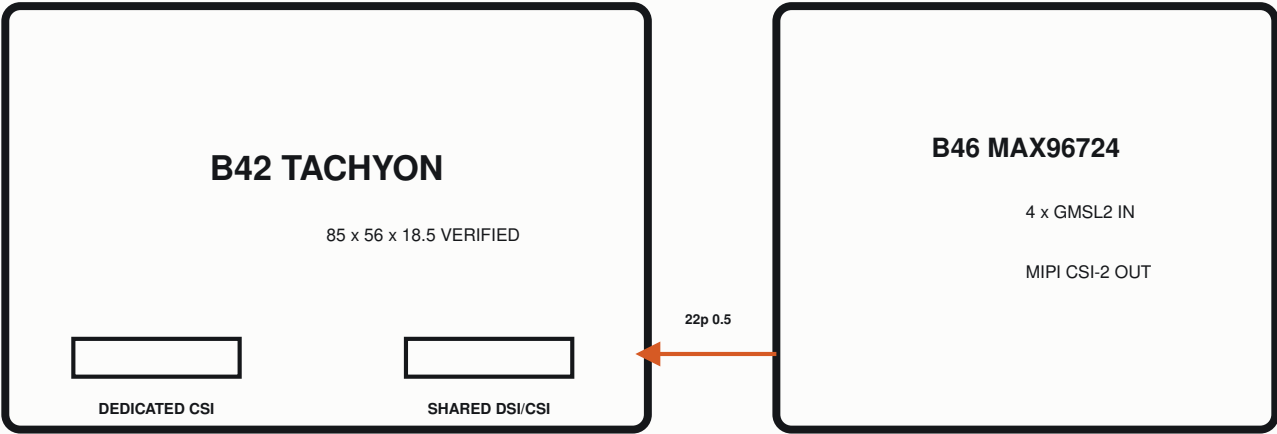
NTS - DO NOT SCALE



SCALE NTS	DATE 2026-08-20	REV B-DRAFT
SHEET 09 OF 12 / S08	STATUS CONCEPT LOCK	DRAWN MANTIS LAB
DWG MT-S08	FORMAT A3 SVG/PDF	TOL AS NOTED

TACHYON + DESERIALIZER CARRIER

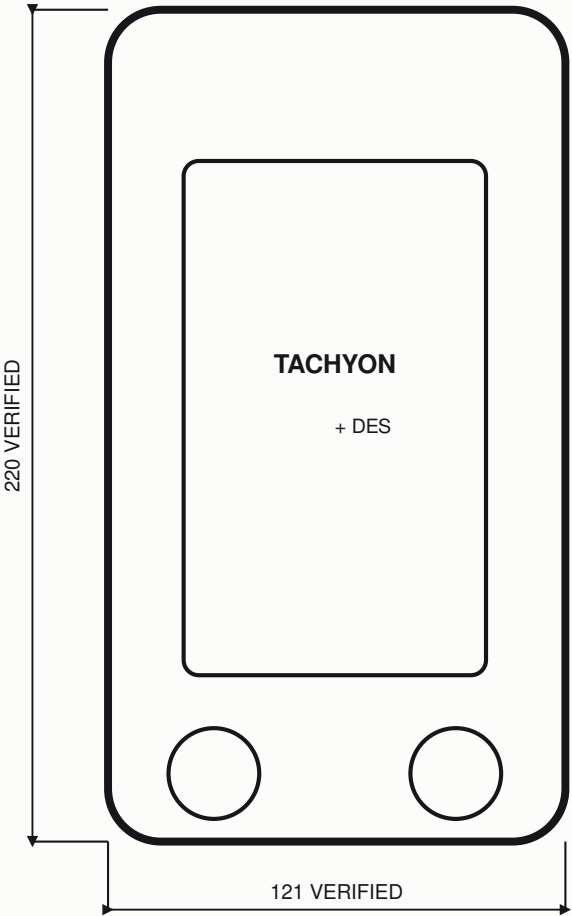
SCALE NTS - DO NOT SCALE



DESERIALIZER CARRIER DIMS / TACHYON DRIVER SUPPORT UNVERIFIED

OPTIONAL M1 EXTERNAL BRICK

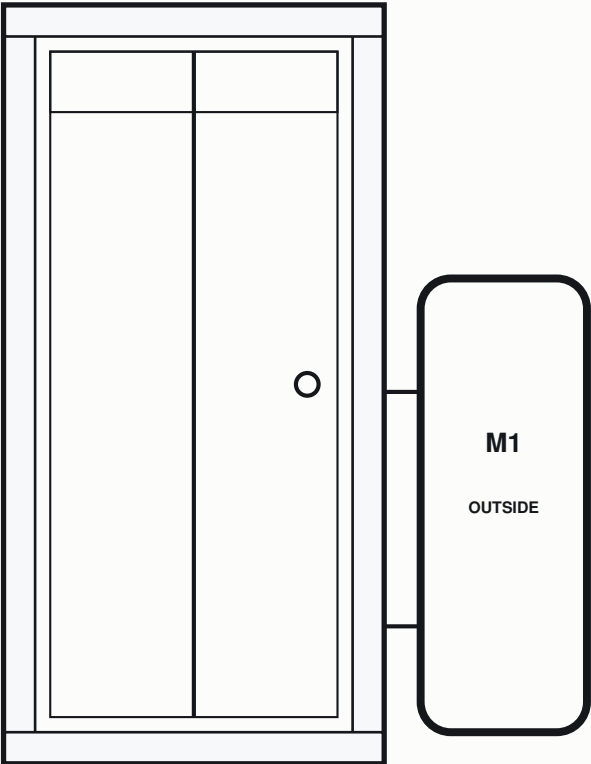
SCALE NTS - DO NOT SCALE



DEPTH 69 VERIFIED
M20 NOT USED FOR MOVING CSI: NO CAMERA TETHER

EXTERNAL MOUNT BOUNDARY

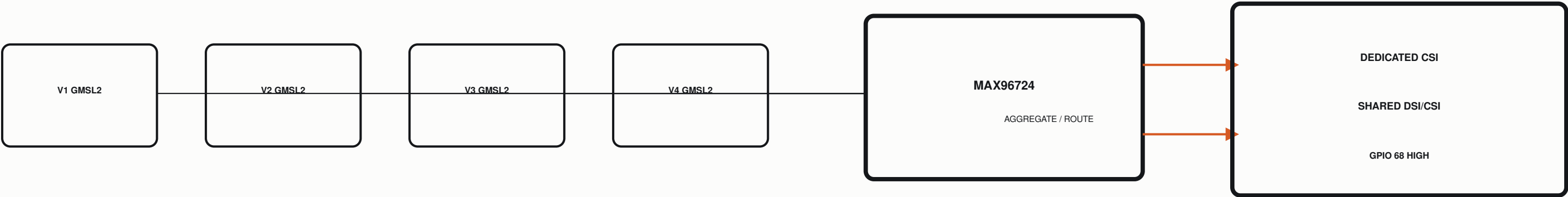
SCALE NTS - DO NOT SCALE



BRICK + SERDES HUB NEVER ENTER WET VOLUME

FOUR VIDEO DOCKS / TWO TACHYON CSI OPTIONS

SCALE NTS - DO NOT SCALE



VERIFIED INTERFACES

Tachyon: dual 4-lane CSI, 22-pin 0.5 mm, up to 2.5 Gbps/lane.
MAX96724: four GMSL2 inputs and CSI-2 outputs; 3/6 Gbps links.

INTEGRATION GATE

Particle does not document MAX96724 support: kernel/DT work is required.
No HAT/CSI pin number is inferred; use Particle and ADI reference CAD.

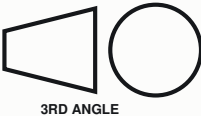
PARTICLE-BASE MANTIS TERRARIUM

SERDES / CSI / PARTICLE BRICK

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE

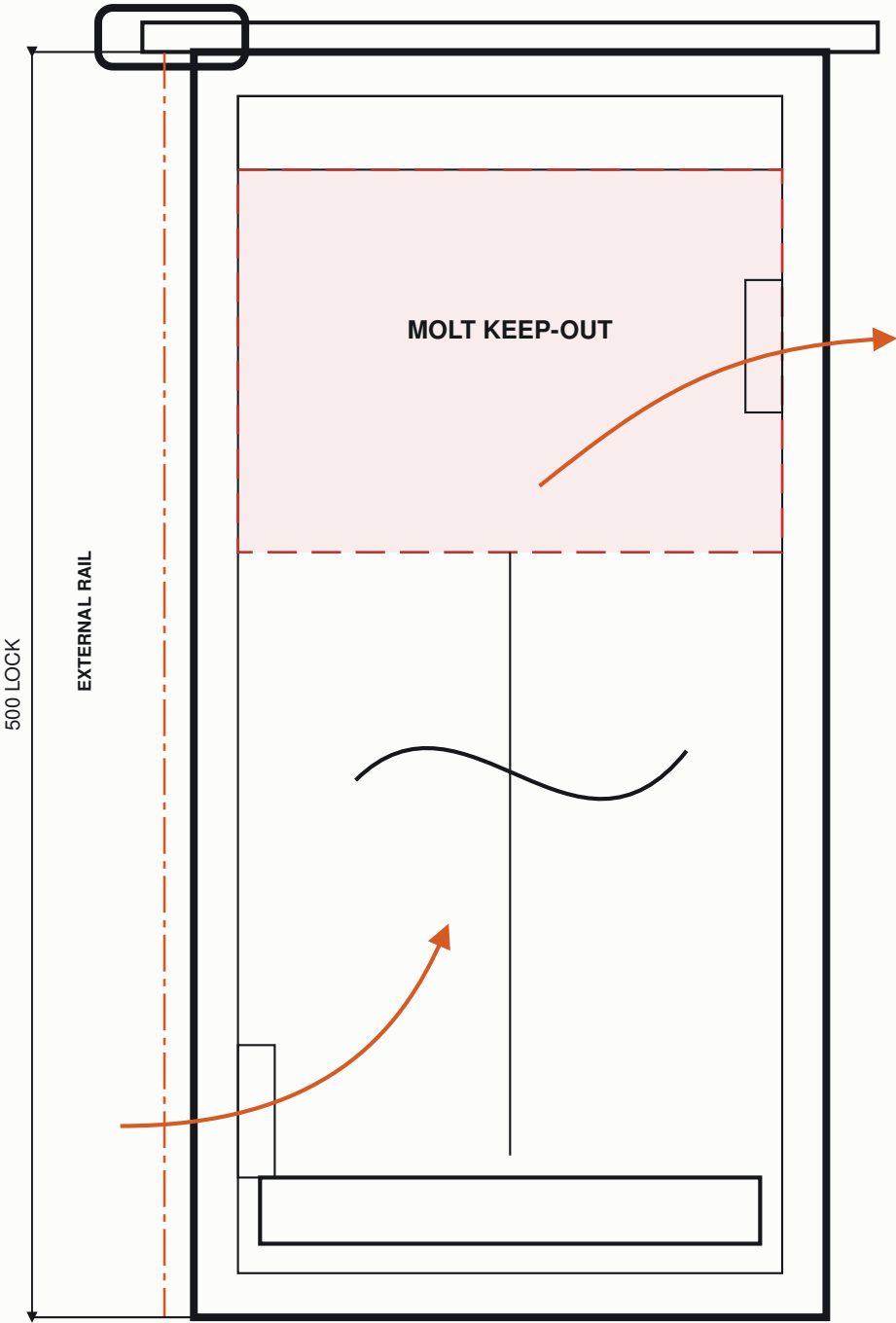


3RD ANGLE

SCALE NTS	DATE 2026-08-20	REV B-DRAFT
SHEET 10 OF 12 / S09	STATUS CONCEPT LOCK	DRAWN MANTIS LAB
DWG MT-S09	FORMAT A3 SVG/PDF	TOL AS NOTED

SECTION THROUGH ANIMAL VOLUME

SCALE NTS - DO NOT SCALE



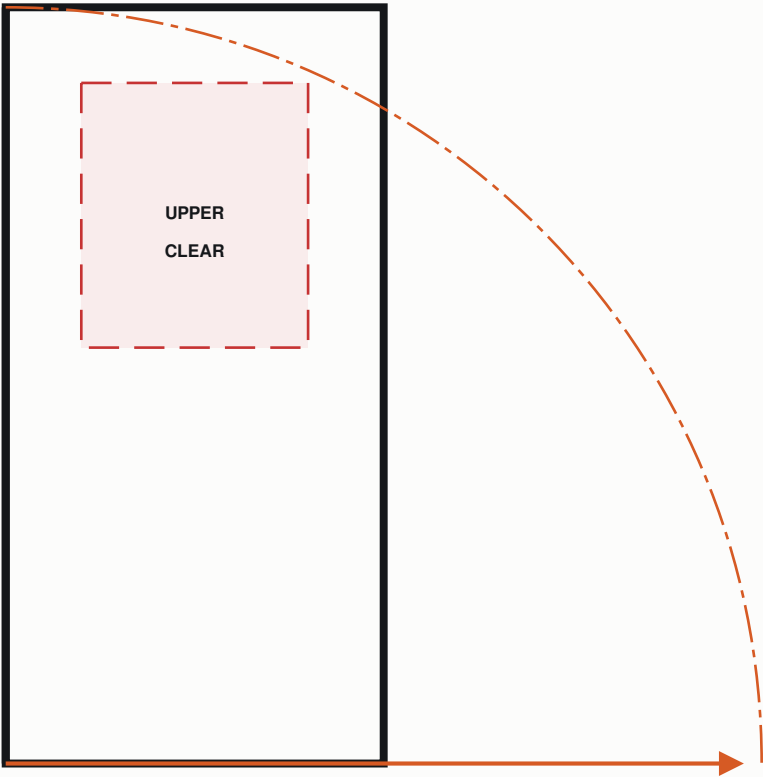
FALSE BOTTOM + 20-30 TRAY

LOW INTAKE -> HIGH EXHAUST

NOMINAL CLEAR 202 W x 202 D x 427 H REF

DOOR SWING / SERVICE ENVELOPE

SCALE NTS - DO NOT SCALE



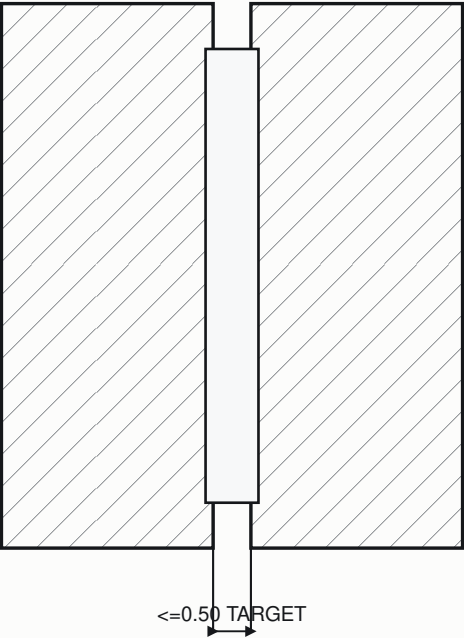
90 deg MIN SERVICE SWING

ANIMAL CHECK

- Solo animal only.
- No taxon inferred: no photo.
- Default adult envelope: 80 mm.
- Interior height must be ≥ 240 mm.
- Ceiling must be textured screen.

DETAIL J - NYMPH-PROOF GAP

SCALE NTS - DO NOT SCALE



LABYRINTH + COMPRESSIBLE WIPER

SCREEN APERTURE ≤ 0.8 LOCK

WET-VOLUME RULE

- No metal mesh.
- No copper / strip / pogo.
- PETG or ASA wet parts.
- Every gland and seam checked.
- Heat-mat pass only; no mains.

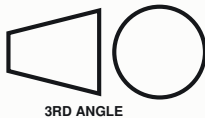
PARTICLE-BASE MANTIS TERRARIUM

HUSBANDRY INTERFERENCE

THEORETICAL - VERIFY AGAINST STEP / FIRST ARTICLE

Units: mm | Projection: THIRD-ANGLE (ASME Y14.3)

NTS - DO NOT SCALE



SCALE
NTS

SHEET
11 OF 12 / S10

DWG
MT-S10

DATE
2026-08-20

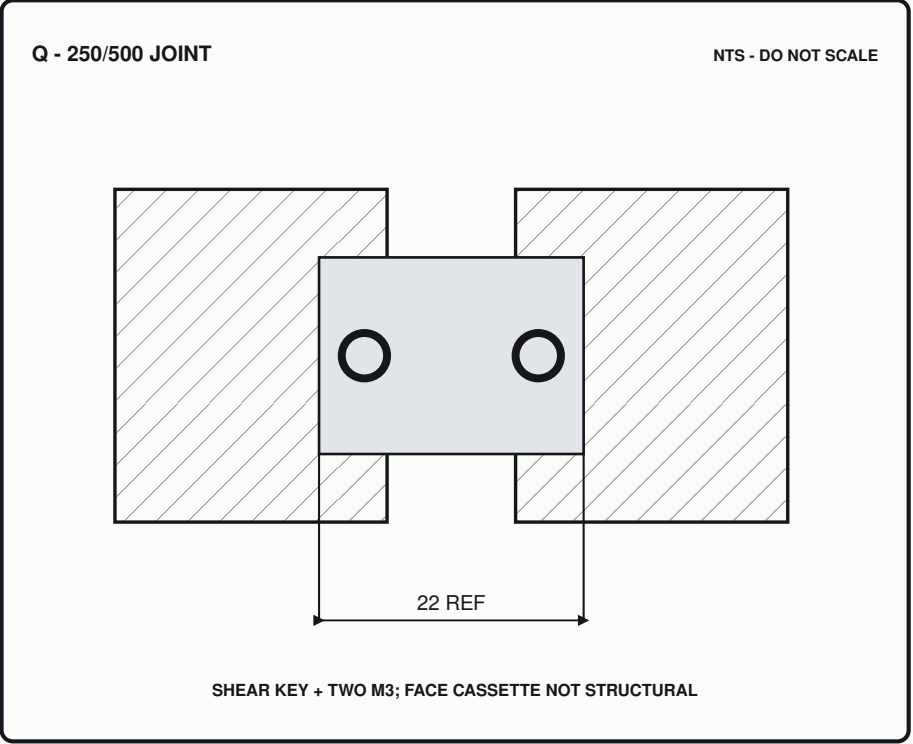
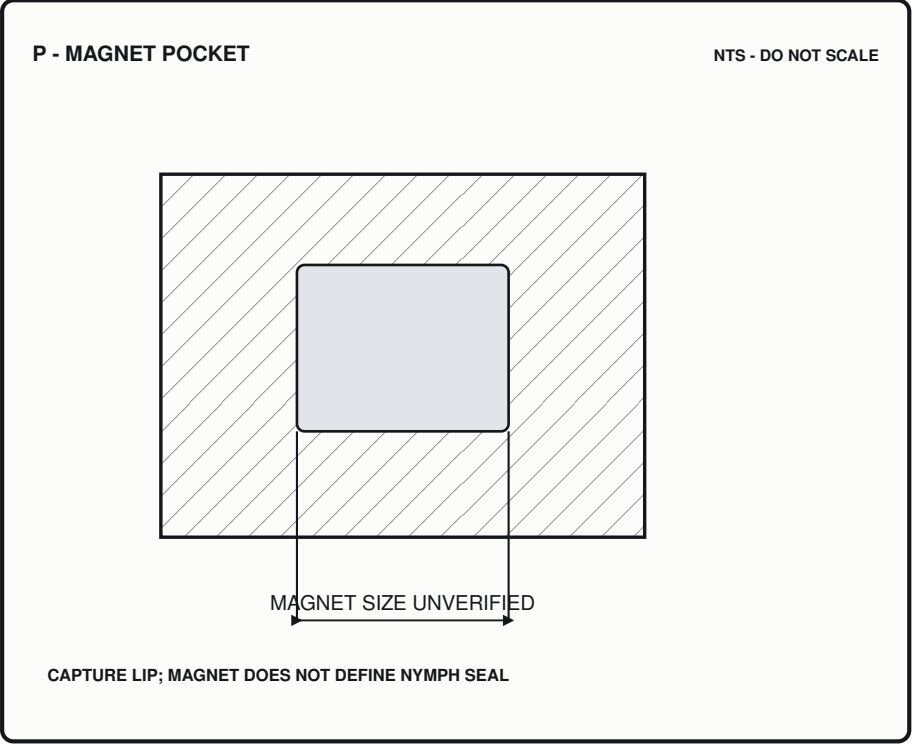
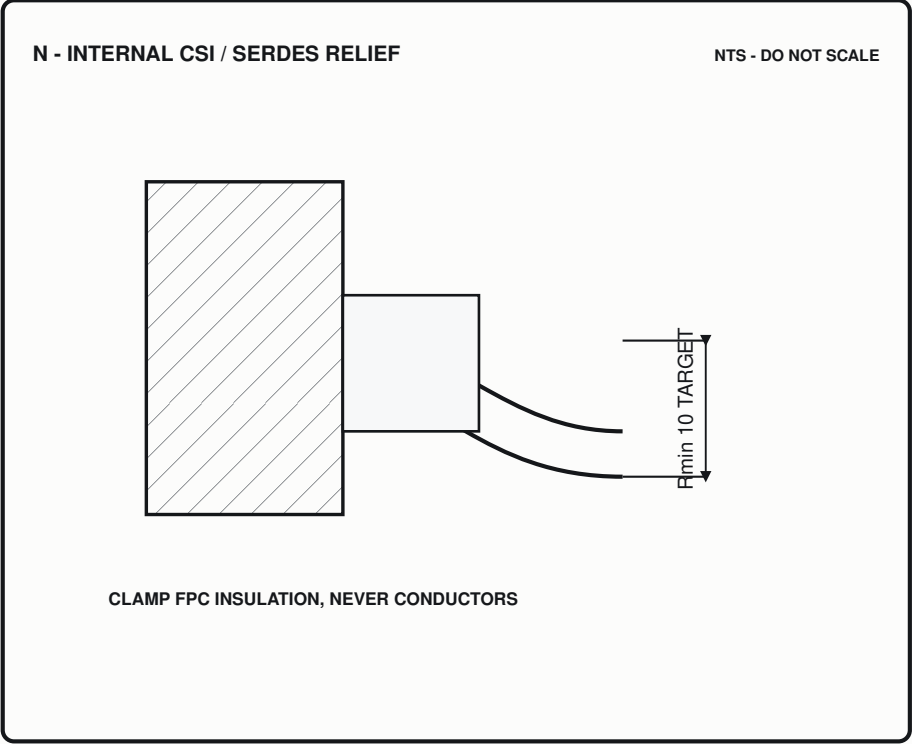
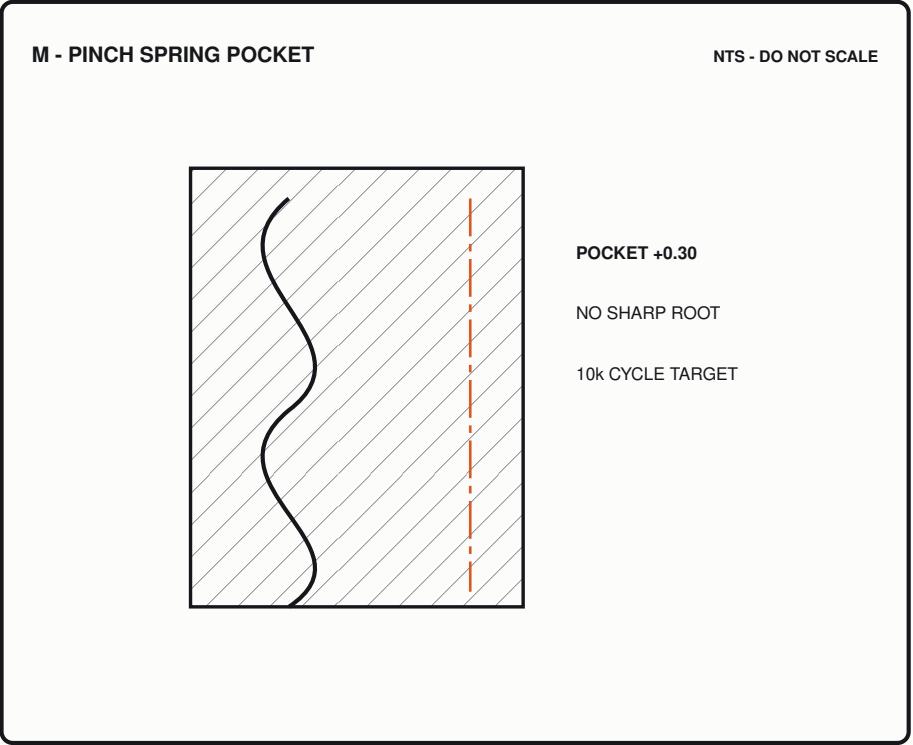
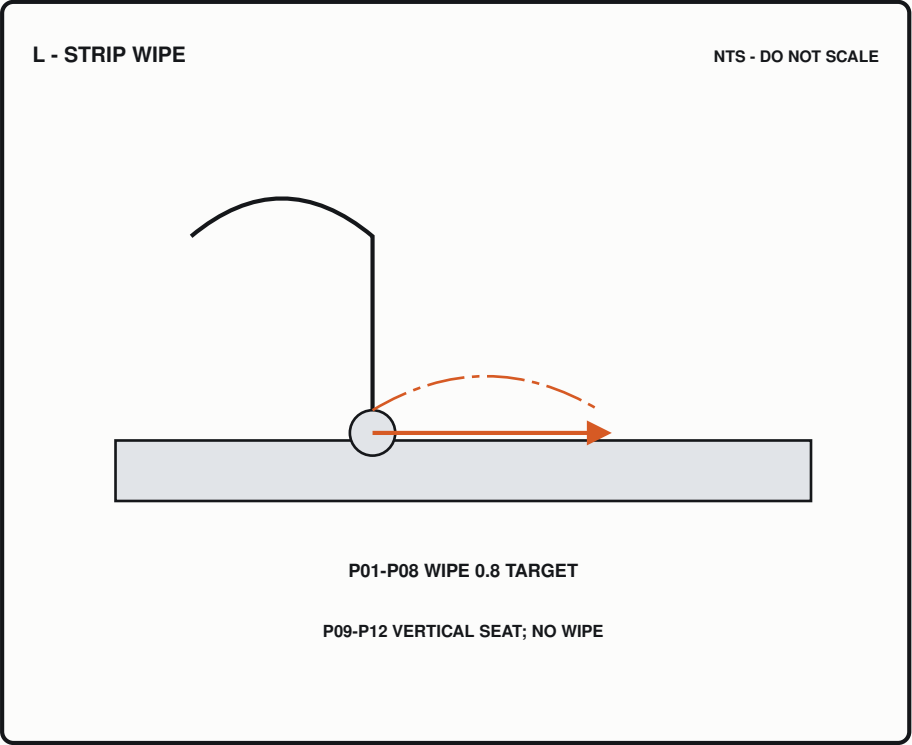
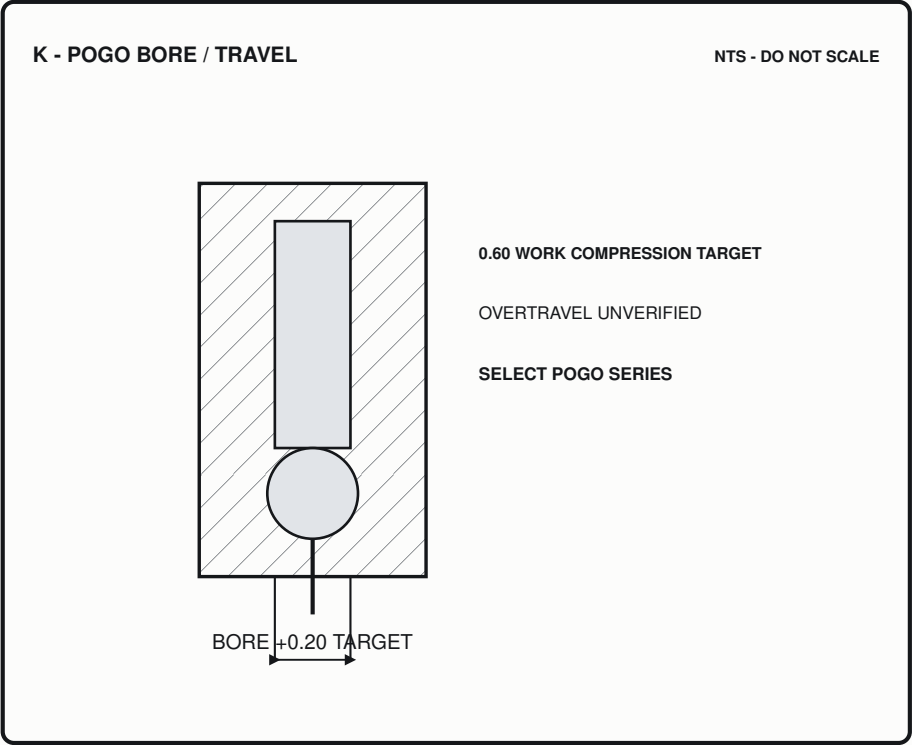
STATUS
CONCEPT LOCK

FORMAT
A3 SVG/PDF

REV
B-DRAFT

DRAWN
MANTIS LAB

TOL
AS NOTED



DETAIL RELEASE RULE

Every value marked TARGET/REF/UNVERIFIED requires coupon or vendor-drawing closure before STEP release.
Critical first articles: pogo bore coupon, strip wipe coupon, latch pull-off coupon, nymph-gap feeler check.